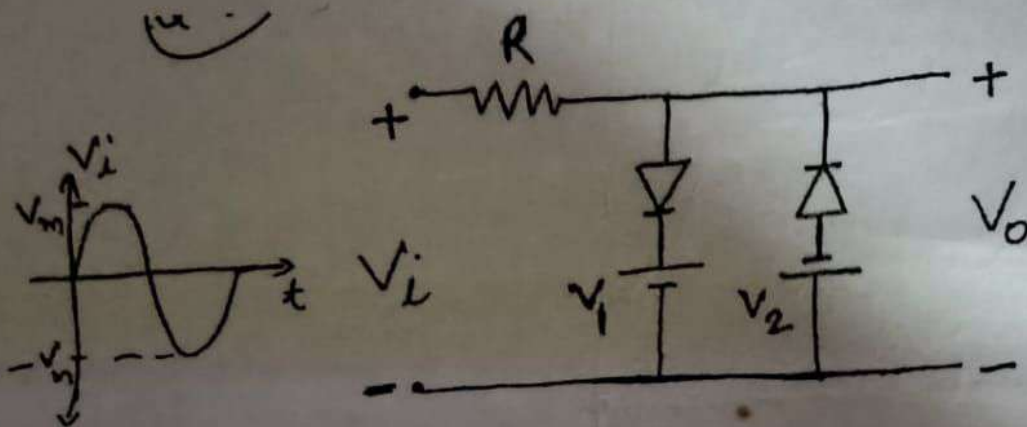


BE Ist Year Test I (CS A& B)
IET-DAVV
Basic Electronics

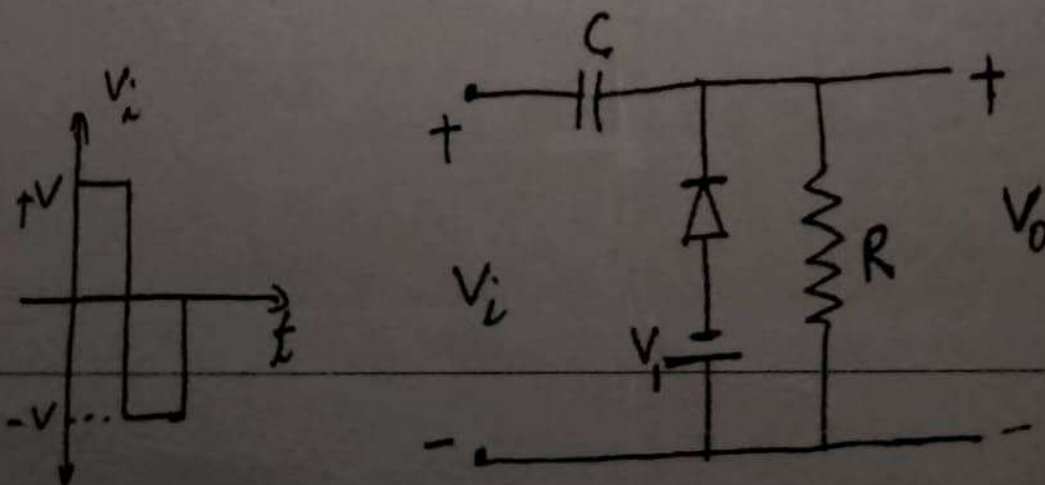
Note: Attempt all questions carrying equal marks.

Q1) When a reverse bias is applied to germanium p-n junction diode, the reverse saturation current at room temperature is $0.3\mu\text{A}$. Determine the current flowing in the diode when 0.15V forward bias is applied at room temperature.

Q2) Draw the output waveform for the following clipper circuit.



Q3) Draw the output waveform of the clamper circuit.



Q4) Design a voltage regulator that will maintain an output voltage of 20 V across a $1\text{-k}\Omega$ load with an input that will vary between 30V and 50V . That is, determine the proper value of R_s and the maximum current I_{ZM} .